PROJECT 2 | SECP3623

CAR RENTAL SYSTEM

Group 1

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Introduction

- Manages **vehicles, customers, booking and payments**
- Customer can:
  - Rent vehicles for specific periods
  - Track booking
  - Make payments (deposits & refunds)

Project Objectives

- Design **NoSQL backend** using MongoDB
- Apply document data modelling principles:
  - Embedding
  - Referencing
  - Arrays
- Demonstrate **CRUD operations**
- Implement **indexing** and **aggregation pipelines**
- Analyse **security considerations** and **design challenges**

Aspect	Embedding	Referencing
Purpose	For tightly related data	For shared data across collections
Data Access	Faster reads (single document query)	Requires joins / $lookup
Data Duplication	Acceptable and common	Avoided
Relationship Type	Strongly dependent	Loosely coupled
Example in System	pickup, return, deposit, refund	booking →customer, booking → vehicle, payment → booking

CRUD Implementation

- Insert Operation (create)
  - **insertOne()**
  - **insertMany()**
- Query Operation (read)
  - find()
  - findOne()
  - Conditional Filtering
    - **$gt**
    - $gte
    - $in
    - **$and**
    - $or
  - **include/exclude field**
- Update Operation
  - **updateOne()**
  - updateMany()
  - **$inc**
  - $push
  - **$pull**
- Delete Operation
  - deleteOne()
  - deleteMany()

Indexing Implementation

- Created indexes on frequently queried fields in the vehicles collection
- Single-field index on vehicle_status
- Compound index on vehicle_location and vehicle_status

Purpose of Indexing

- Improve data retrieval speed
- Reduce full collection scans
- Support faster vehicle availability searches

Query Performance Impact

- Indexed queries return results more efficiently
- Improved scalability as data volume increases
- Enhances overall system responsiveness

Aggregation Pipeline

Total revenue for “paid” bookings grouped by payment method

- $match
- $group

Total payment amount within a time range, grouped by payment status

- $match
- $group

Bookings with non paid payment statuses

- $match
- $lookup
- $unwind
- project

Deposit refund status count and total deposit amount

- $group

Customers with forfeited deposits

- $match
- $lookup
- $unwind
- project

Most rented vehicle class

- $match
- $lookup
- $unwind
- $group
- $sort

The top 3 vehicle make owned

- $group
- $sort
- $limit

Sorting demonstrations

Purpose of Sorting

- Improve data readability
- Support efficient data analysis
- Enhance system usability and reporting

Example

- Sorting Customers’ Contact Information by Name (Ascending)
- Sorting Paid Payment by Amount (Descending)

Performance Improvements

by Advanced Operations

Indexing

- Reduces full collection scans
- Enables faster data retrieval on frequently queried fields

Aggregation Pipelines

- Processes and summarizes data within MongoDB
- Reduces unnecessary data transfer and improves analytics efficiency

Sorting

- Organizes query results in meaningful order
- Improves usability after filtering operations

Team Member Contribution

Team Member	Role and Responsibilities
Teh Ru Qian	• Provide the project overview, domain description, objectives • Design the NoSQL data model
Tan Yi Ya	• Implement the basic MongoDB CRUD operations • Summarize the project outcomes
Woo Cheng Shuan	• Design and demonstrate indexing strategies • Analyze query performance and optimizing retrieval in MongoDB
Lam Yoke Yu	• Create and demonstrate aggregation pipelines to perform complex analytics • Generate operational insights from the database
Goe Jie Ying	• Implement sorting operations • Discuss security measures for sensitive data and analyze the limitations of MongoDB

THANK YOU FOR LISTENING!